

health SURGERY

make a cut and with radical improvements in medical imaging, access parts of the body as never before. "Surgery may not yet feature *Star Trek*-ian techniques, but in some regards they are on the way," says Dr Naresh Trehan, the cardiologist who pioneered robotic surgery of the heart in the country in 2002.

Check out the surge of "first-ever" surgeries all across the country: in August, nerve electrodes enabled a ventilator patient to breath normally at AIIMS; in July, doctors at Jaslok Hospital in Mumbai performed a dual pacemaker surgery; in June, AIIMS achieved a landmark in robotic chest surgery. In March, at Bangalore's Narayana Hrudayalaya, "Asia's first artificial heart" was implanted. In January, tissue glue was used to fix up an intraocular lens in Chennai. Last year, orthopaedics replaced a patient's kneecap totally in Delhi, while artificial spinal discs were put in Mumbai and Bangalore. In 2006, Asia's first "Brain Suite"—the most advanced technology to treat brain tumours—was launched at the Institute of Neuroscience, Max Hospitals, in Delhi. In 2003, almost like science fiction, Drs G.V. Rao and Nageshwar Reddy at the Asian Institute of Gastroenterology, Hyderabad, removed an appendix through the mouth—the first in the world.

Driven by the rising health-care demands and spending power of India's affluent generation, medical technology looks set to enter a golden age. A new Ernst & Young study predicts 15-20 per cent growth for the Indian medical equipment market, slated to reach to \$5 billion by 2012, from \$2 billion now. "That doesn't necessarily mean treatment will be cheaper," says Kumar, professor of surgery at AIIMS. "But it promises to deliver more for the money. Almost certainly, it will be less dangerous and painful for patients."

From February this year, Ela Srivastava, 21, of Kanpur started getting periods of extreme weakness, when doors were too weighty to open, stairs too difficult to climb, food too hard to chew, eyes too droopy to focus. She suffered the ultimate onslaught—respiratory failure and paralysis—in May. It was a rare neuromuscular disorder and demanded a fierce chest surgery with large incisions sawed straight through the breastbone. But in June, when Kumar operated on Srivastava at AIIMS—one of the five robotic surgery centres in India—no rib or muscle was torn asunder, no nerve crushed and the patient could walk on her own within 48 hours.

THE NEW-AGE OT

Rapidly advancing technologies are changing operating rooms at astonishing speed and daring surgeons to keep up

ENDOVISION

System used for the robot. Has high-resolution digital camera, direct recording on computer and high-resolution monitors for 3-D visualisation.

ELECTROCAUTERY

The crude old machine for cutting tissue and stopping blood loss during surgery is now a sophisticated electrothermal system

ENDOVISION

For endoscopic surgery. Digitalised X-ray images and endoscopy findings can be directly transmitted to the surgeon's screen.

ROBOT

Tiny cameras attached to the four arms are inserted into the patient. Each arm has surgical instrument attached to it. The operation is done through pencil-sized punctures, as surgeon commands via a console.